2022 JDP – AI Detection and Causality Inferences for Fab Operations

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Date: 2022/08/11

Outline

- Thoughs on log parsing
- Surveyed Paper
  - On Automatic Parsing of Log Records
    - ICSE 2021 (#15 conference), #citations = 2
  - Prefix-Graph: A Versatile Log Parsing Approach Merging Prefix Tree with Probabilistic Graph
    - ICDE, 2021, unsupervised

Review - 張敬

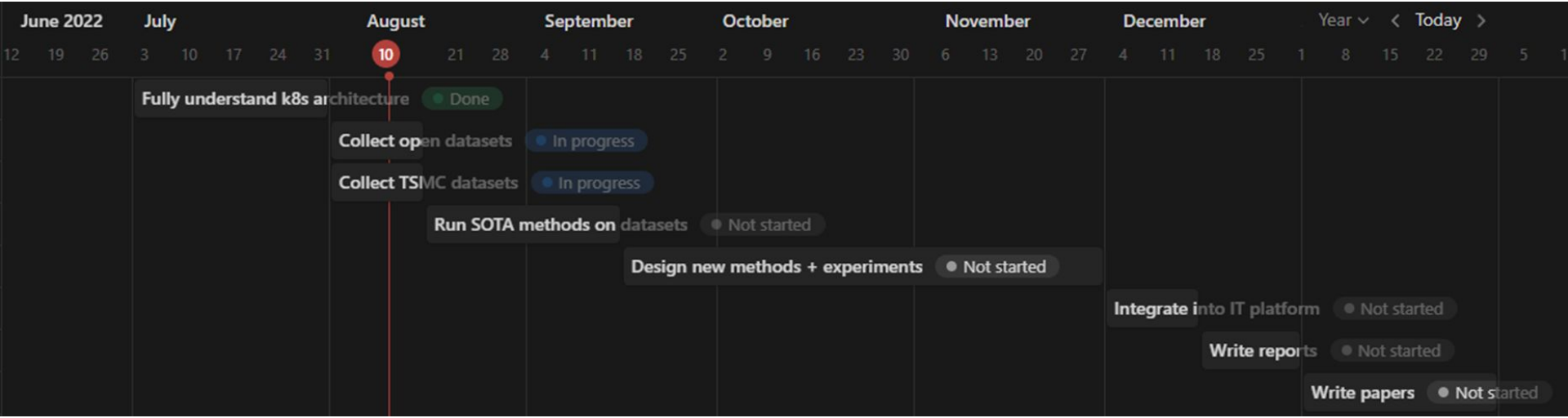
- Done in the past two weeks
  - Understand kubernetes, Helm, Istio
    - Train Ticket with k8s + isio (using minikube to create k8s cluster)
  - Thoughs on log parsing
- To do
  - Produce all 14 possible anomalies from Train Ticket
  - Unsupervised POS tagging papers

Review - 正明

- Done in the past two weeks
  - Log parsing post-processing
    - Merge the different cluster but similar template
- To do
  - Apply the method from paper to TSMC data

Internship plan (from NYCU)

Name	Status	Date
Fully understand k8s architecture	Done	2022/07/01 → 2022/07/31
Collect open datasets	In progress	2022/08/01 → 2022/08/15
Collect TSMC datasets	In progress	2022/08/01 → 2022/08/15
Run SOTA methods on datasets	Not started	2022/08/16 → 2022/09/15
Design new methods + experiments	Not started	2022/09/16 → 2022/11/30
Integrate into IT platform	Not started	2022/12/01 → 2022/12/15
Write reports	Not started	2022/12/16 → 2022/12/31
Write papers	Not started	2023/01/01 → 2023/01/31



Problem Settings

Supervised or Unsupervised	Unsupervised
Offline or Online	Online
Data type	Metric + Log + Trace + Changelog
Data source	System log (kubernetes-related) Application log
Others	1. Multiple sources a. time mismatch b. log sparsity issue (error message) 2. Log parsing 3. Causality/Interpretability/Explainability

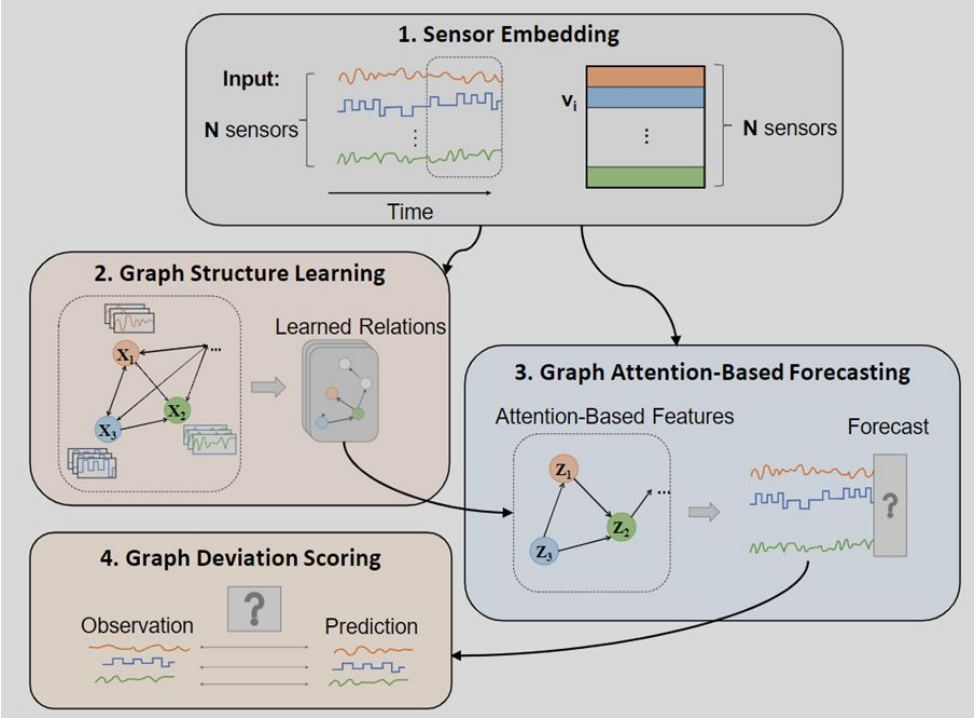
Keywords: micro services, causal graph, root cause analysis, causal inference

Problem Settings

Worth Implementing	Name	Training manner	Problem Setting	Input	Output
☐	TSMC JDP 2022	Unsupervised	Root Cause Analysis	Metric, Log, Trace, Changelog	root cause services
☑	DeepTraLog	Unsupervised	Anomaly Detection	Log, Trace	An anomaly score for each trace
☑	MicroHECL	Unsupervised	Root Cause Analysis	Metric, Trace	Locations and types of the root cause services
☐	CloudRCA	Unsupervised	Root Cause Analysis	Metric, Log	
☐	GDN	Unsupervised	Anomaly Detection	Metric	
☐	NeuralLog	Unsupervised	Anomaly Detection	Log	
☐	GLAD-PAW	Unsupervised	Anomaly Detection	Log	

Single source – Metric

- GDN (AAAI 2021)
  - GNN-based MTS AD
  - If using Graph Deviation Scoring, output will be anomalous label for each timestamp
- Darts
  - Nbeats, TCN, TFT, NHiTs



Time Series Made Easy in Python

Darts

Not specified for microservices

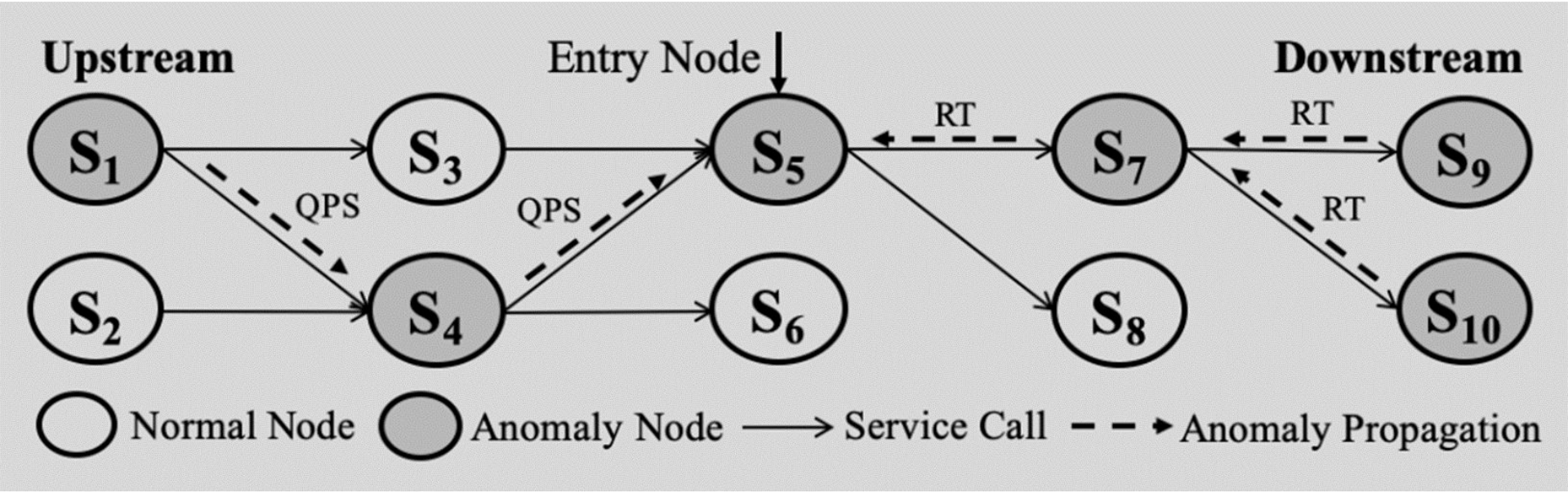
Single source – Log

- NeuralLog
  - Log AD without log parsing
- GLAD-PAW
  - Log AD with GNN
- LogAnomaly
  - …

~~Multiple sources – Metric + Trace~~

Single source – Trace

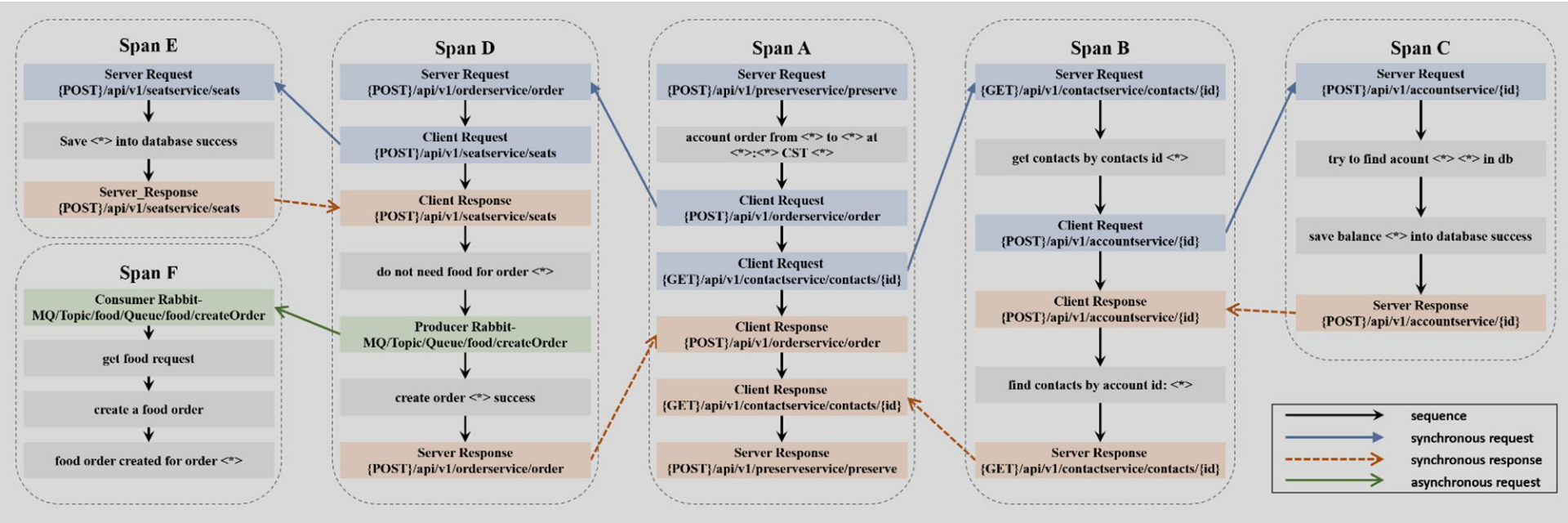
- MicroHECL



"Trace" is not just about the topology/edges. We can have some metric/KPI on the edges.

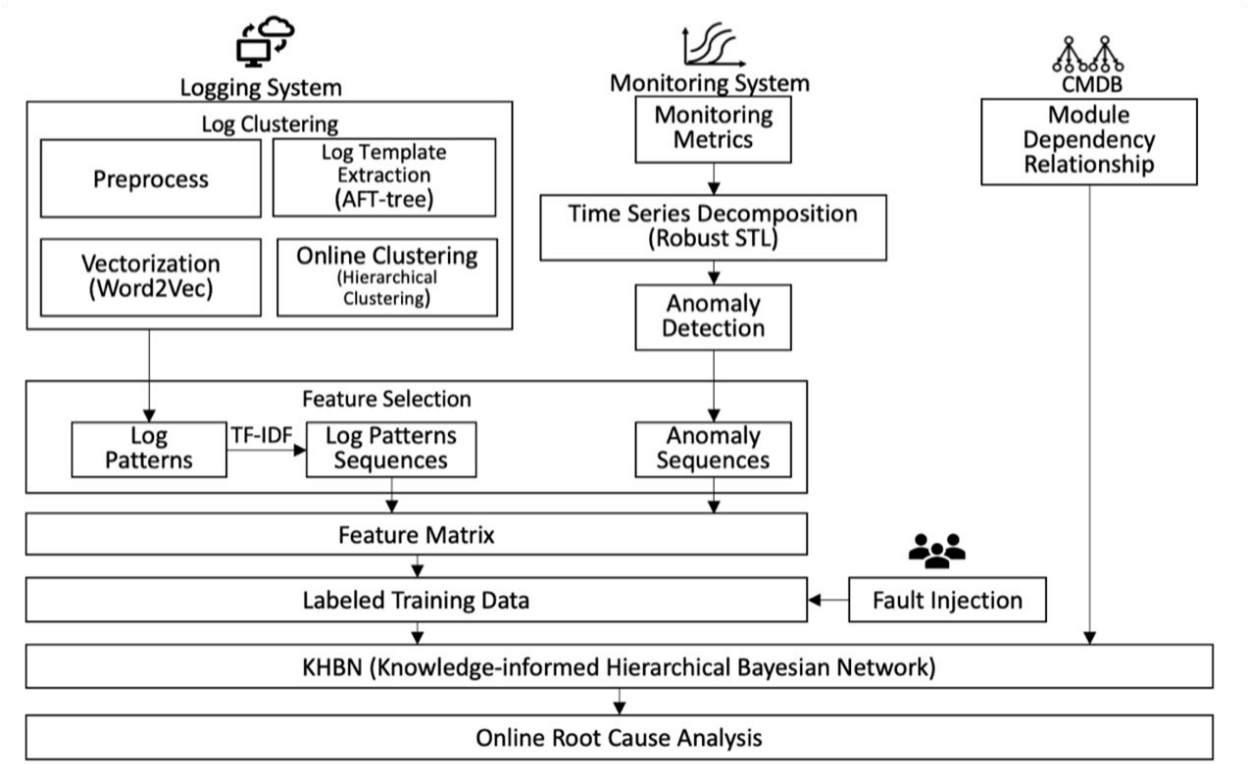
Multiple sources – Log + Trace

- DeepTraLog



Multiple sources – Metric + Log

- CloudRCA



3 Selling points for my own paper

- Combine 3 data sources
  - Log, Trace, Metric
  - Change “Changelog” to “Metric”, like TSMC did

- GNN with RCA

- Self-supervised learning (with GNN)
  - Contrastive learning

Tsinghua NetMan Lab – NetManAIOps

Autonomizing IT Operations

Open Source!

- TraceAnomaly
  - ISSRE 2020
- TraceRCA
  - IWQoS 2021
- DejaVu
  - Actionable and Interpretable Fault Localization for Recurring Failures in Online Service Systems
  - FSE 2022
- CIRCA
  - Causal Inference-Based Root Cause Analysis for Online Service Systems with Intervention Recognition
  - KDD 2022

Implementation

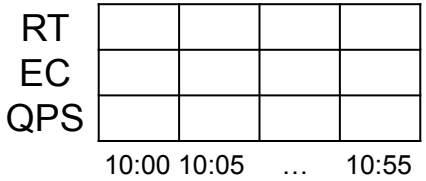
- MicroHECL with TSMC dataset
- DeepTraLog with Train Ticket

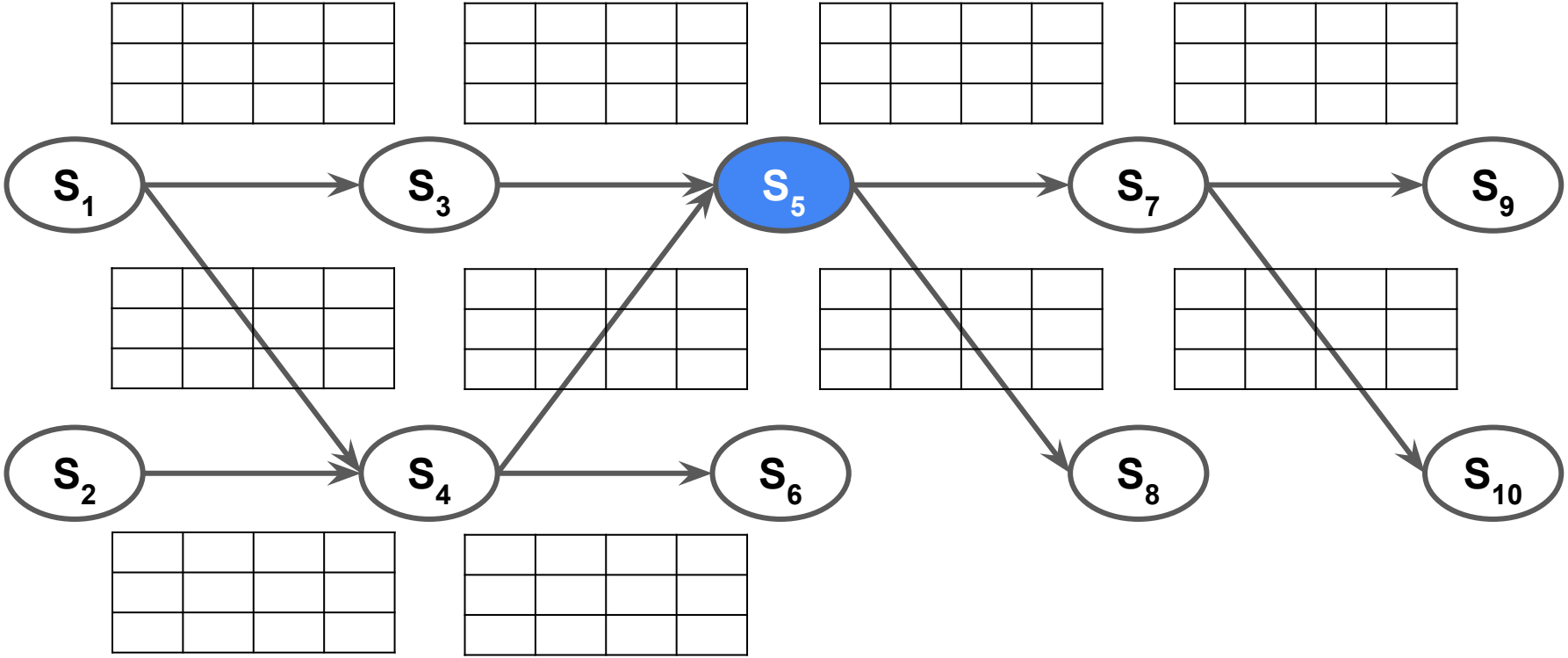
MicroHECL

Trigger point: having availability issue (business metric)

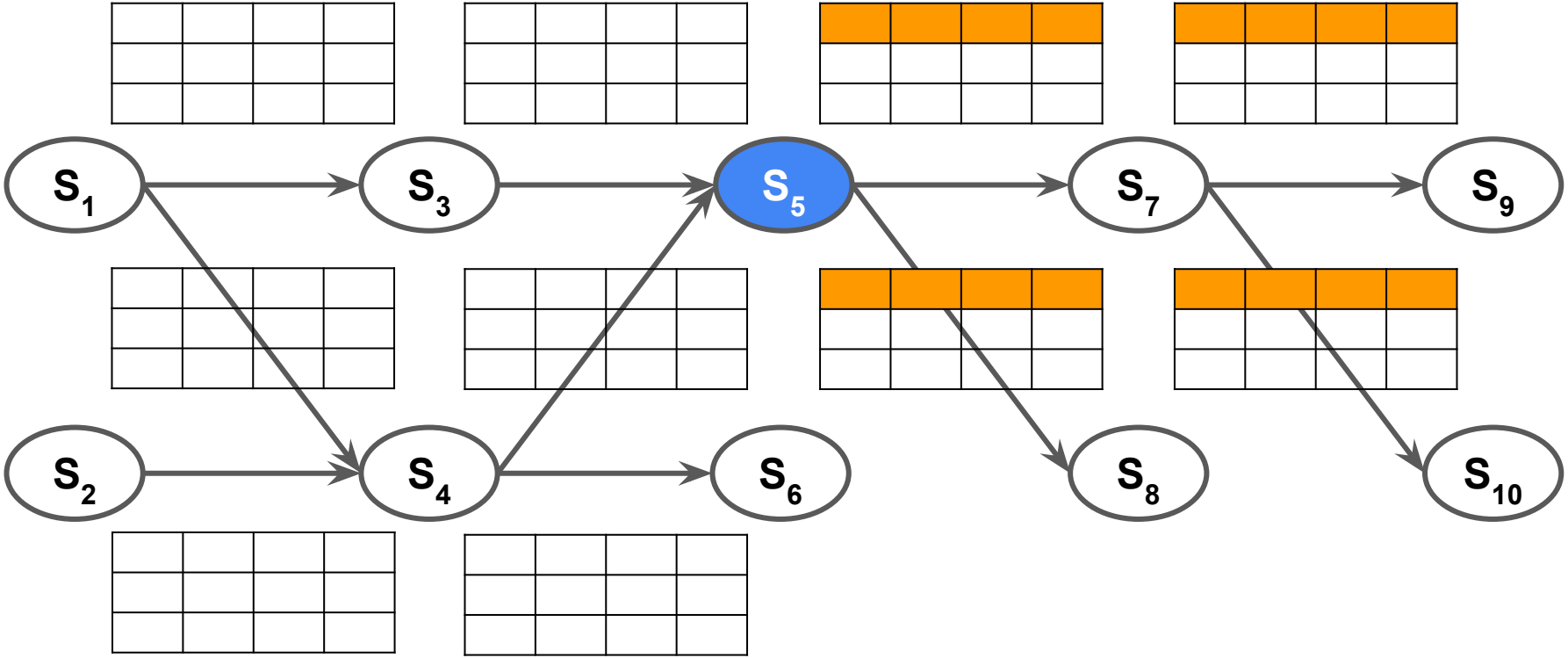
1. Service Call Graph Construction (in the latest time window, e.g., the last 30 min)
2. Anomaly Propagation Chain Analysis
3. Candidate Root Cause Ranking

1. Service Call Graph Construction

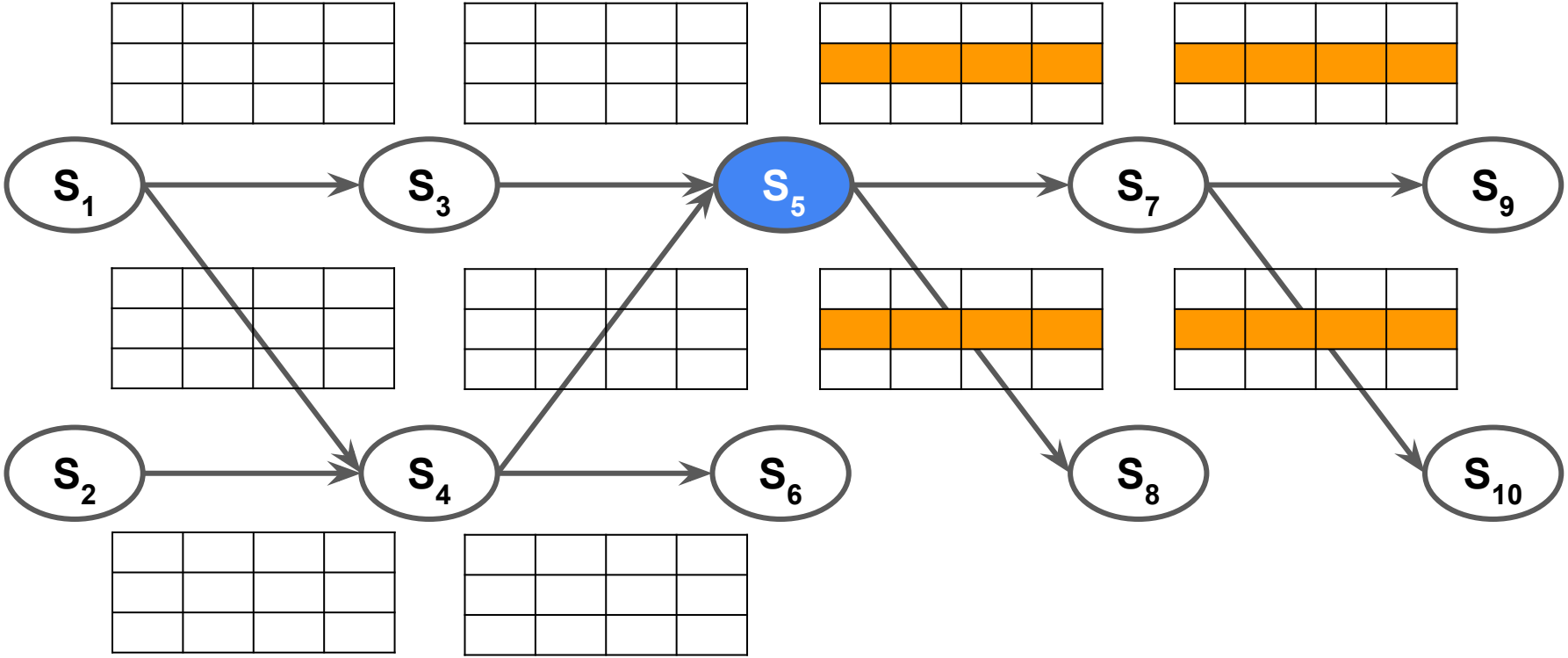




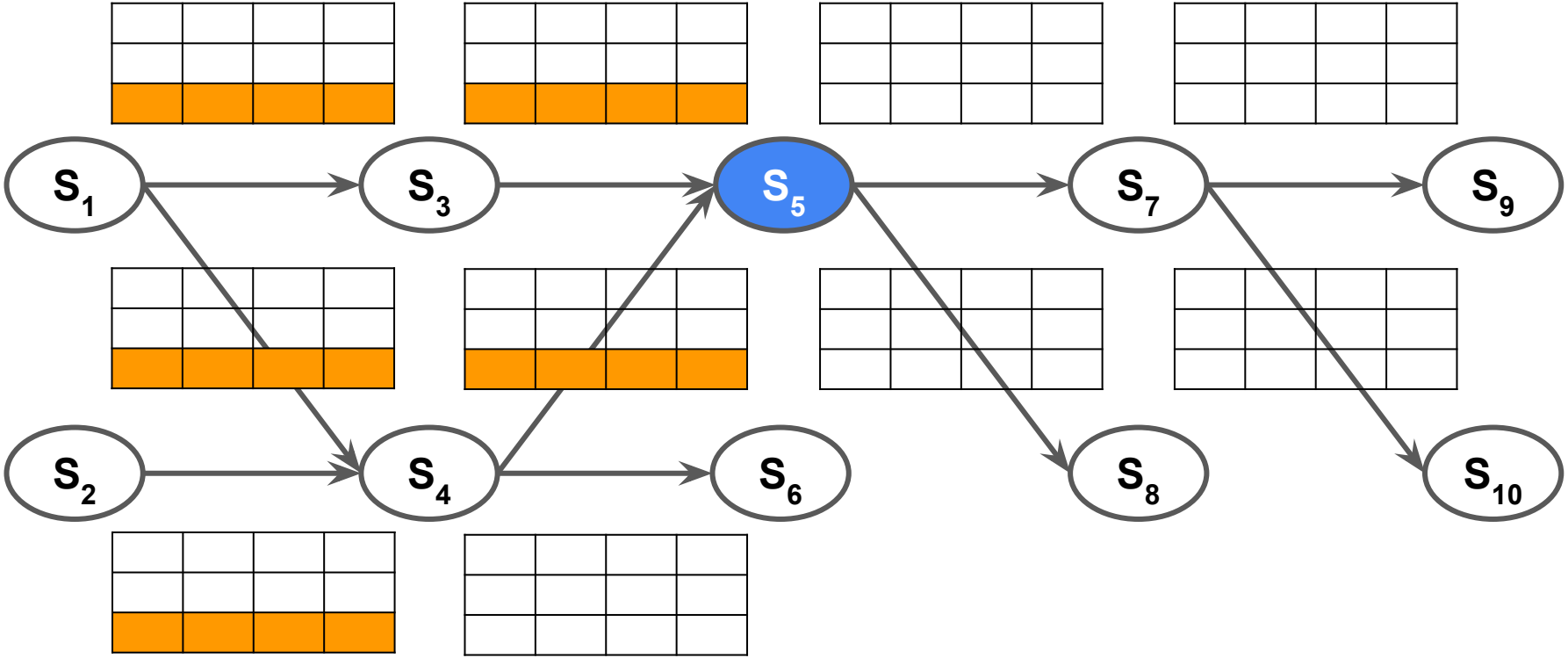
2. Anomaly Propagation Chain Analysis (RT)



2. Anomaly Propagation Chain Analysis (EC)



2. Anomaly Propagation Chain Analysis (QPS)



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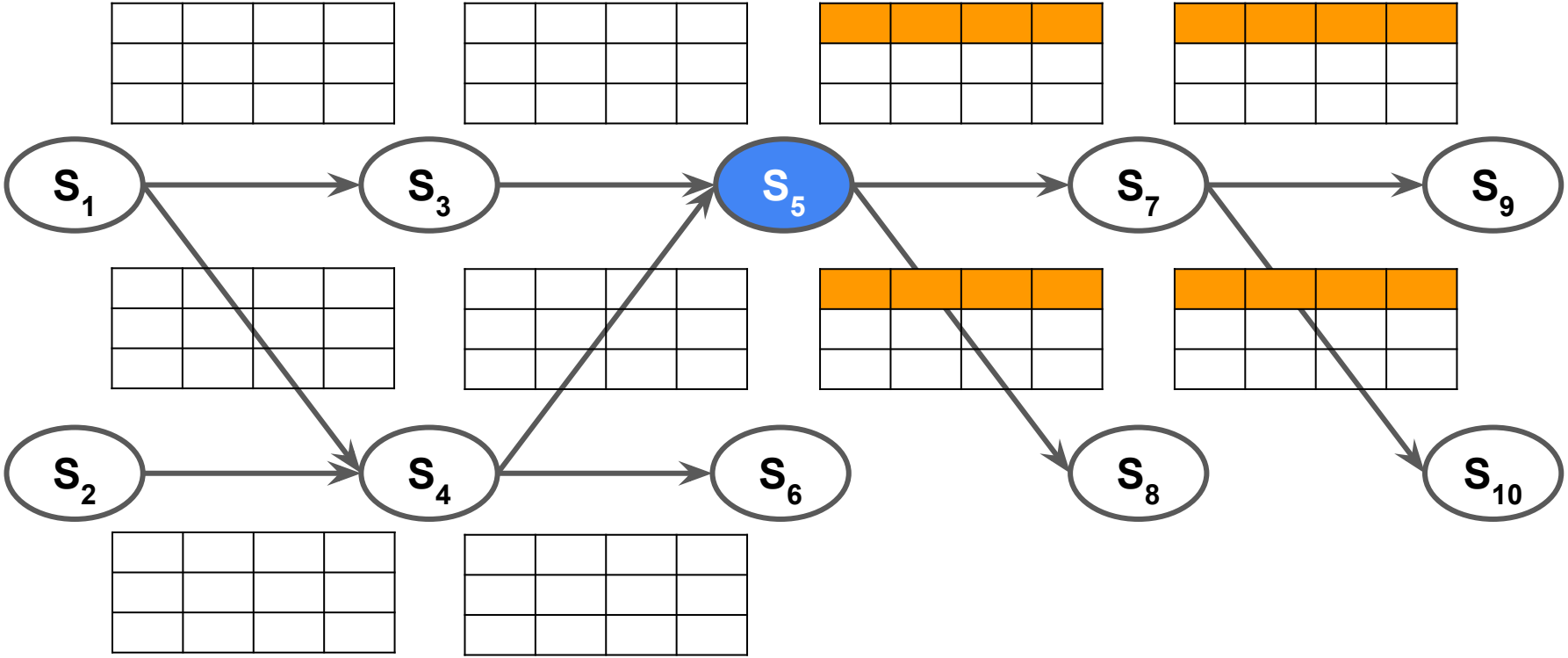

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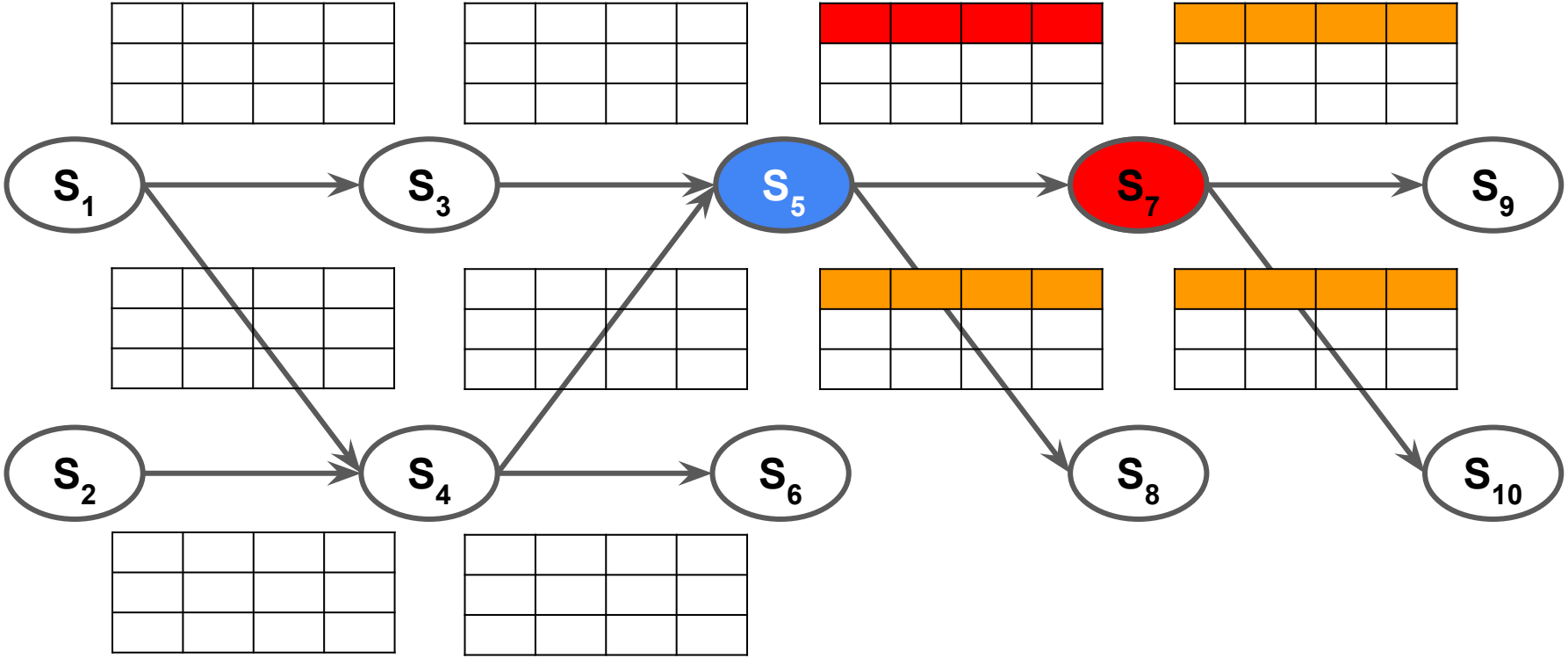

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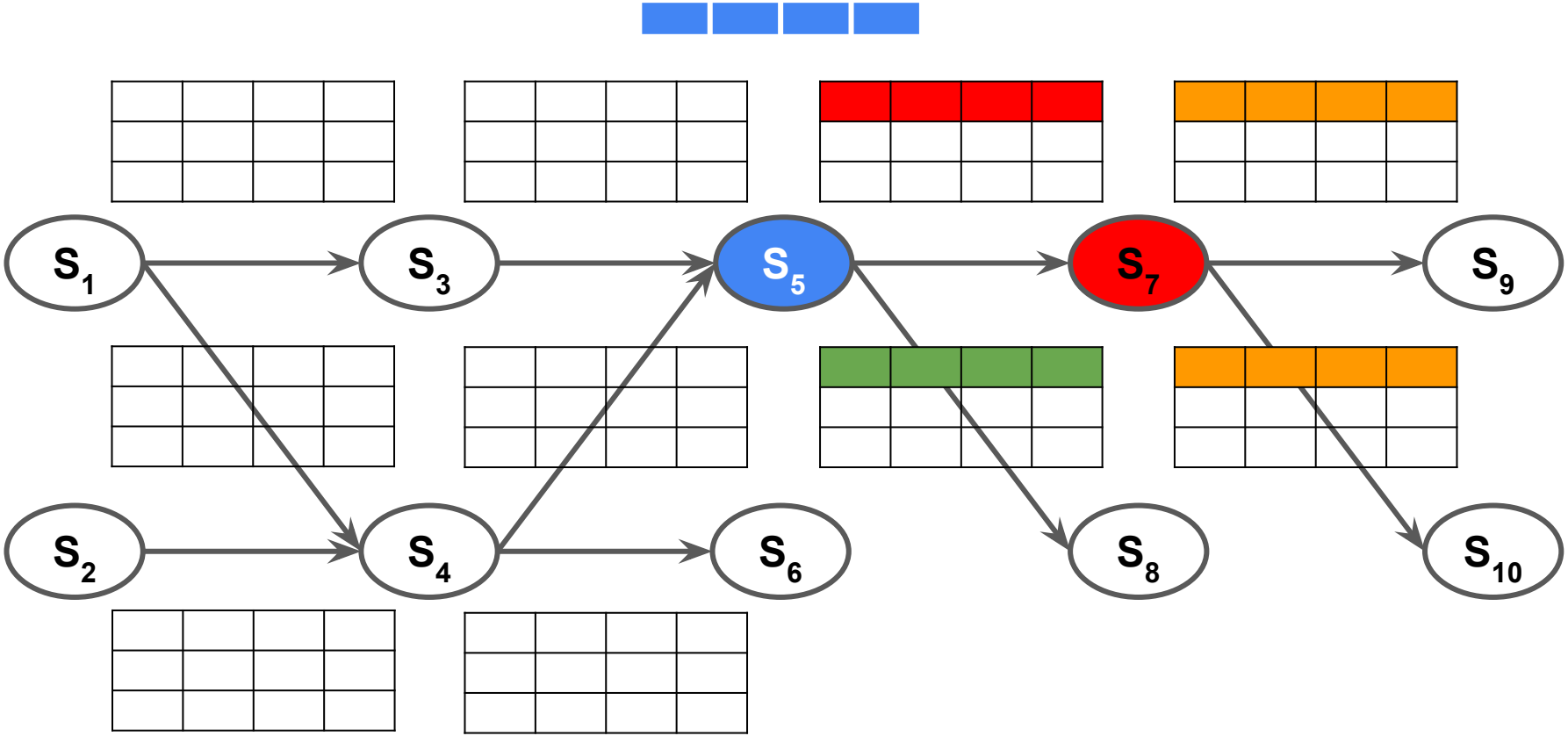
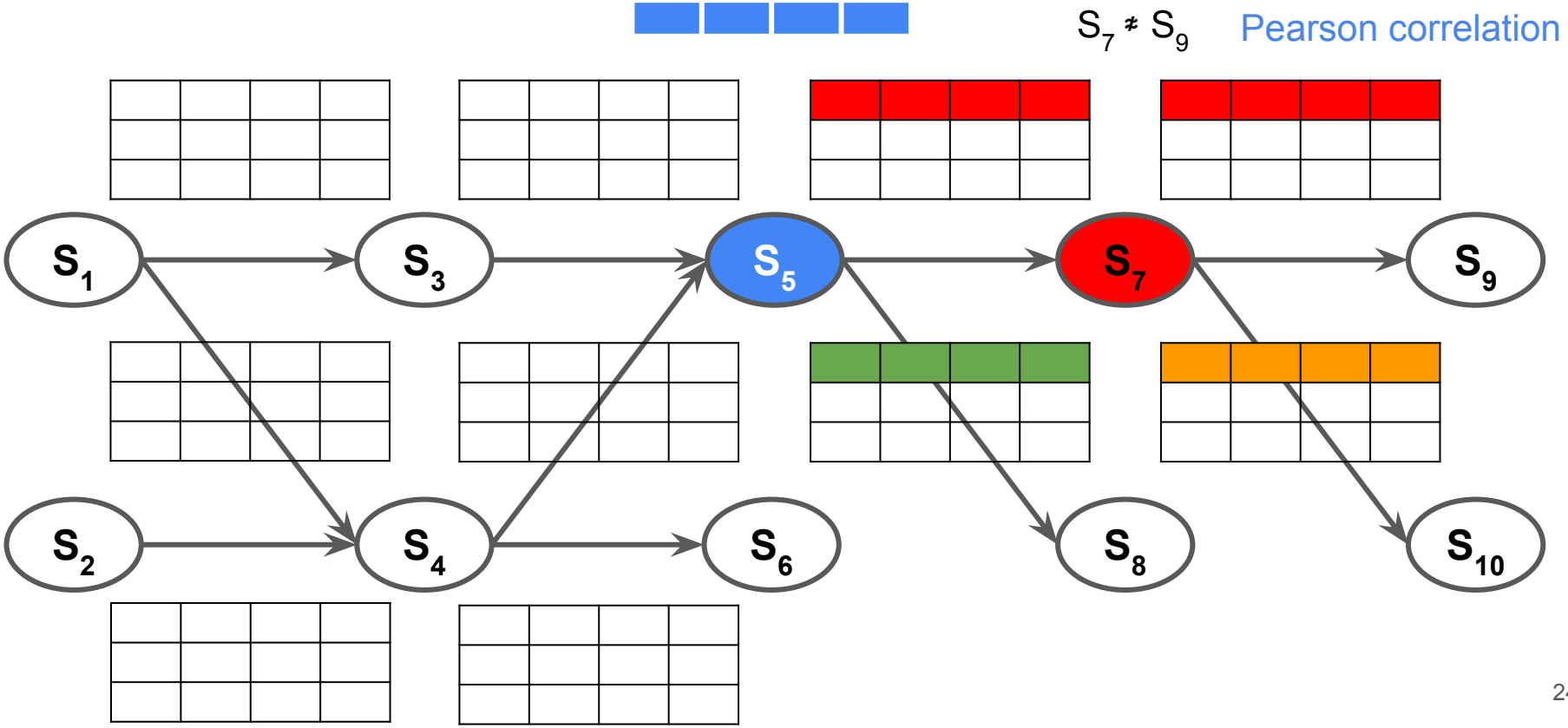


$S_5 \approx S_7$ ✗ No need

2. Anomaly Propagation Chain Analysis (RT)

$S_7 \not\approx S_9$ Pearson correlation



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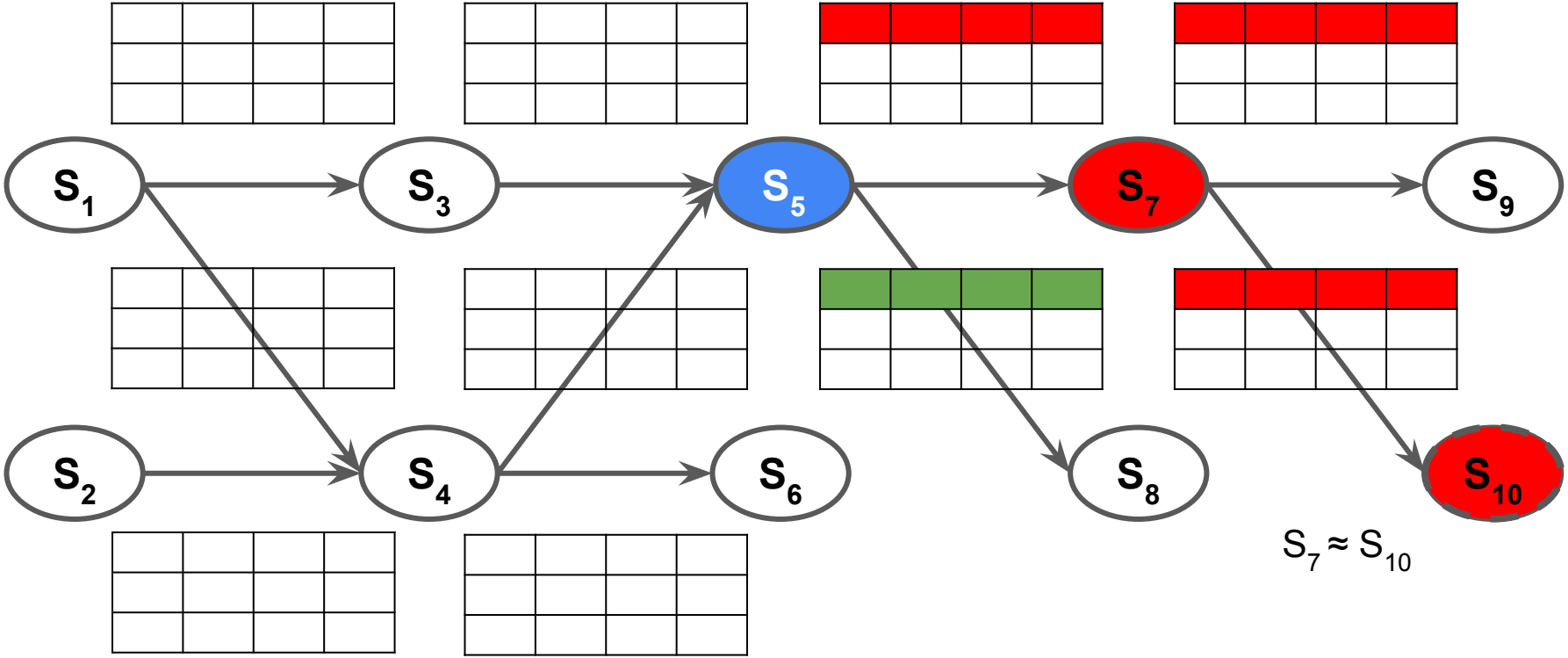

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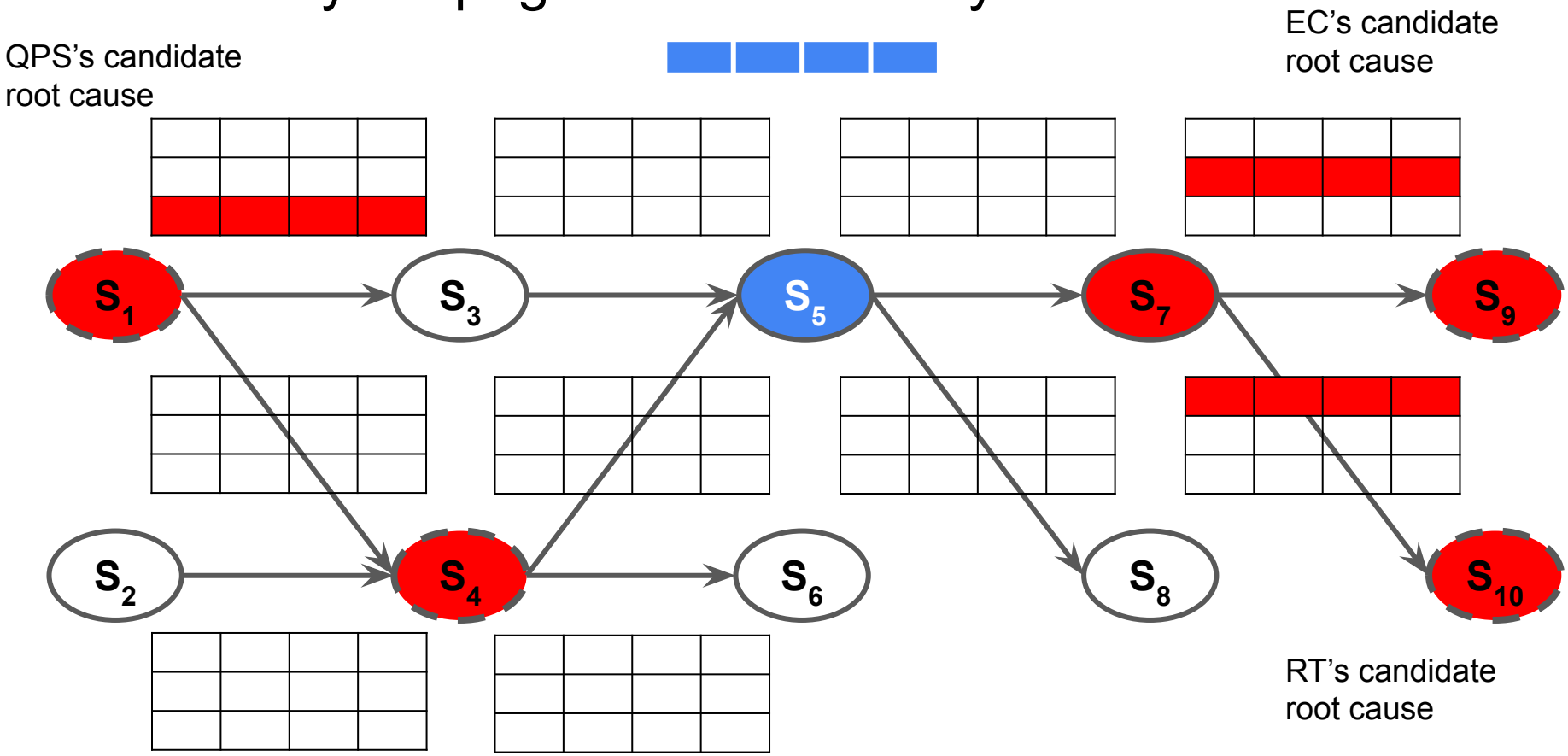
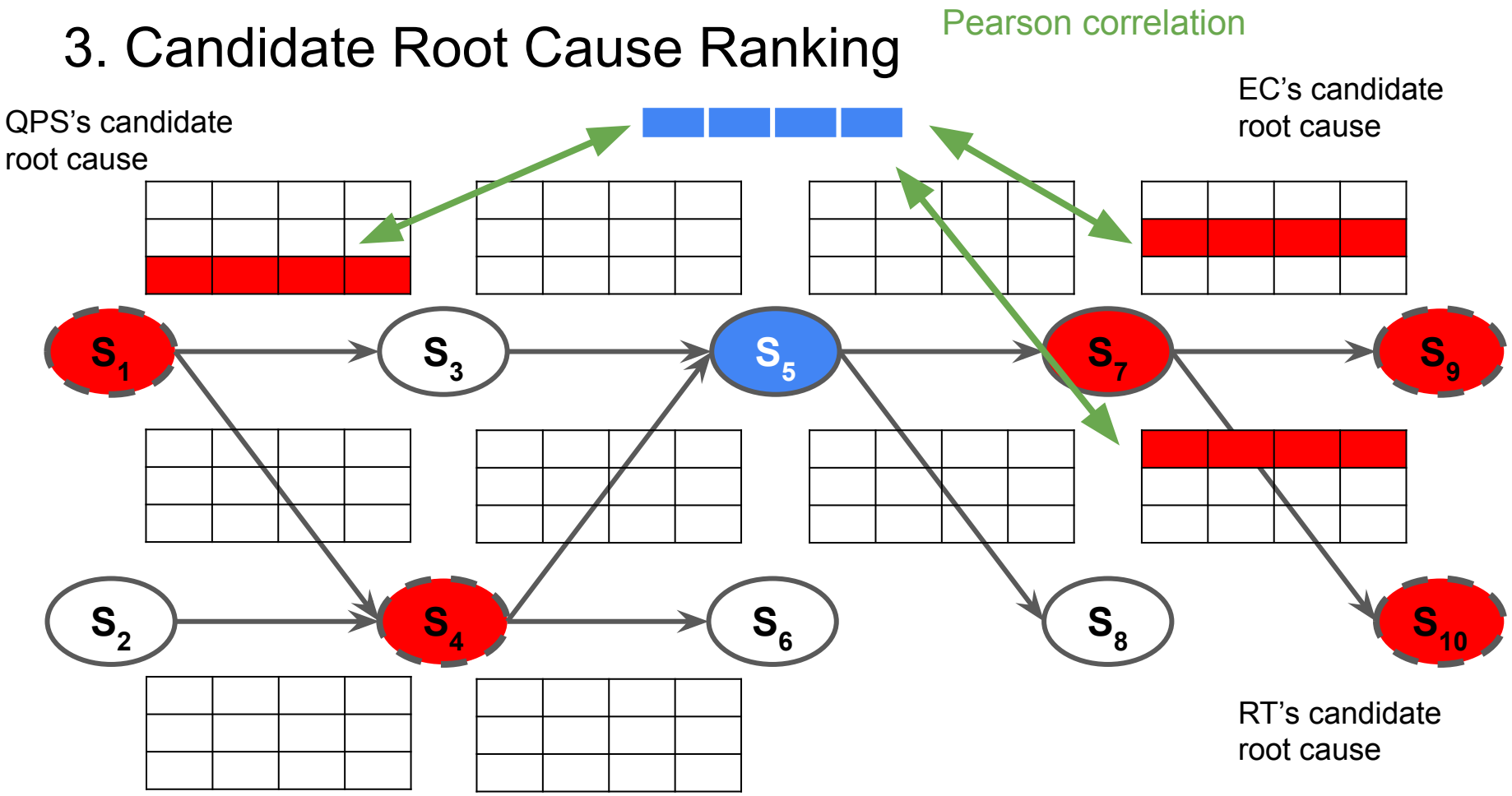


3. Candidate Root Cause Ranking



MicroHECL with TSMC dataset

- get_metric.py
  - Find the business metric (開單成功率)
  - Initial anomalous service
  - Problem:
    i. 開單成功率的sample rate從SAO看起來很大，是每小時一筆，但是從 CY的圖片可以觀察到，其實開單率的sample rate應該要很小，像是每 5分鐘就會有一筆資料
    ii. 從SAO撈出來的每小時的資料，如果我們用 service的名稱來篩選開單成功率的話，會有很多筆records。我不確定我這邊是要取哪一筆。

MicroHECL with TSMC dataset

- get_trace.py
  - Find the topology and quality metrics around the given initial anomalous service
  - Problem:
    i. 不確定是否能直接用service name來跟metric資料做結合

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Current Problem

- Undermasking
  - Similar template diff cluster / Different template same cluster

Method

- First, calculate the similarity of two log templates
  - Use pretrained model **Sentence-BERT** to embed log template
  - Then, calculate the cosine of two embedding vectors as the similarity
- Then, calculate the similarity of the appearance time of two log templates
  - The size of the time vector is 288 (Per 5 minutes in 1 day)
  - As the same as first step, calculate the cosine of two time vectors as the similarity

Pros

- Successfully to merge some different cluster but their are same event (as we can check the result by auto-feedback)
- Find the same event but doesn’t be discovered by feedback
  - Ex: cluster11473, cluster14870

Cons

- Different appearance time but similar raw log (only different parameters)
  - Ex: cluster15689, cluster16044

Discussion

- How to define the performance
- Recall or precision